Electromagnetic Waves

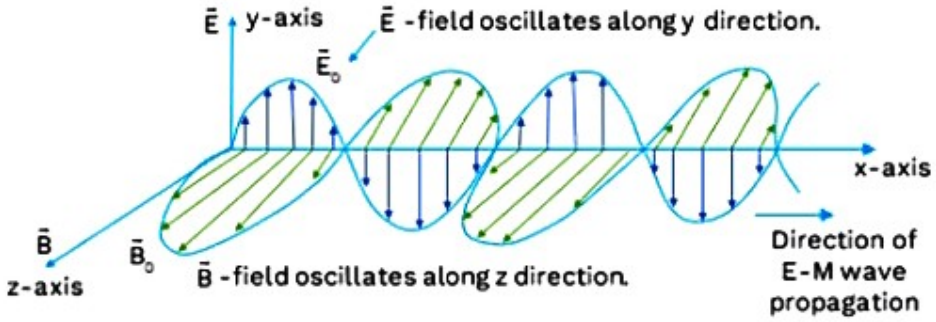
Electromagnetic Waves

Introduction :- According to Maxwell, an accelerated charge produces a sinusoidal time-varying magnetic field which in turn produces a sinusoidal time-varying electric field. The two fields so produce mutually perpendicular time-varying electric and magnetic fields. They constitute electromagnetic waves that can propagate through empty space.

Important Points About Electromagnetic waves.

1. The source of E-M wave is accelerated charge.
2. The E-M wave is transverse in nature.
3. Direction of propagation of EM wave is perpendicular to both electric field and magnetic field.
4. Direction of propagation of EM wave is along $\vec{E} \times \vec{B}$.

Graphical Representation :-

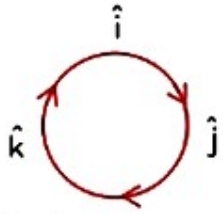


$$\vec{E} = \{E_0 \sin(\omega t - kx + \phi_0)\}\hat{j}$$

$$\vec{B} = \{B_0 \sin(\omega t - kx + \phi_0)\}\hat{k}$$

$\vec{p} = \dfrac{(\vec{E} \times \vec{B})}{\mu_0}$ "The Poynting vector represents the directional energy flux of an electromagnetic wave."

$\vec{E}$ =Electric field oscillates along y-axis.
$\vec{B}$ = magnetic field Oscillates along z-axis.
$\vec{P} = \vec{E} \times \vec{H} \rightarrow \hat{j} \times \hat{k} = \hat{i}$



Electromagnetic Wave Propagates along the x-axis in this given situation.

Concept Of Displacement Current :-

Definition :- Displacement current is that current which comes into play in the region in which the electric field Changing with time, and due to this, the electric flux is changing with time.

$$I_D = \epsilon_0 \frac{d\phi_E}{dt}$$

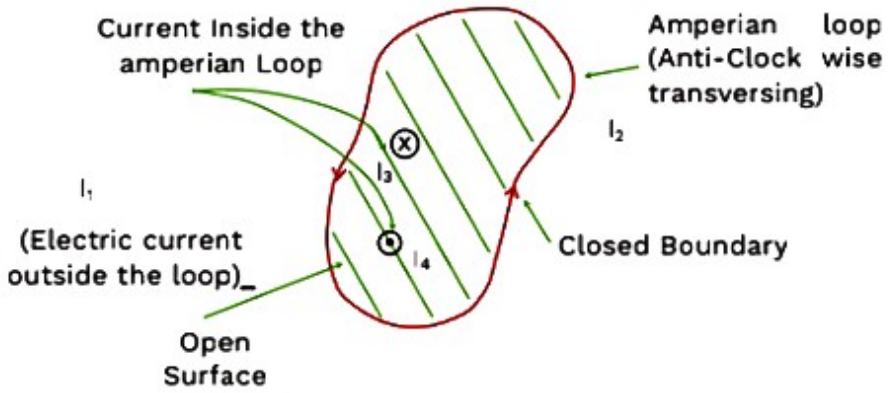
Where : ϕ_E = Electric Flux .

I_D = displacement current.

Inconsistency of Ampere's Circuital Law and Displacement-Current:-

According To ampere's Circuital Law

$$\oint \vec{B} \cdot \overline{d\ell} = \mu_0 I_{in}$$



$$\oint \vec{B} \cdot \overline{d\ell} = \mu_0 I_{in}$$

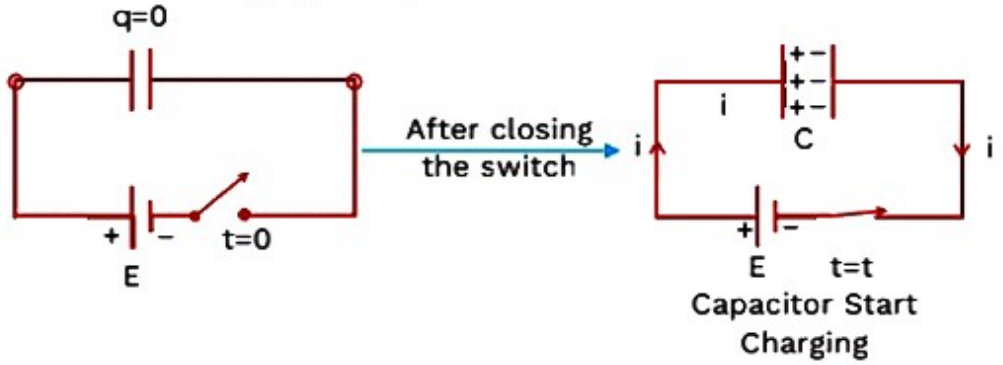
$$\oint \vec{B} \cdot \overline{d\ell} = \mu (I_4 - I_3)$$

Circle Represent Closed Boundary

Where;

$I_{in} = I_4 - I_3$ = Inclosed current crossing the surface

Now consider a Case of charging of Capacitor :-



Now we want to calculate $\vec{B}$ at a distance r from wire.

Case-1 :- If we select Circular Amperian Loop as shown in ther figure:-

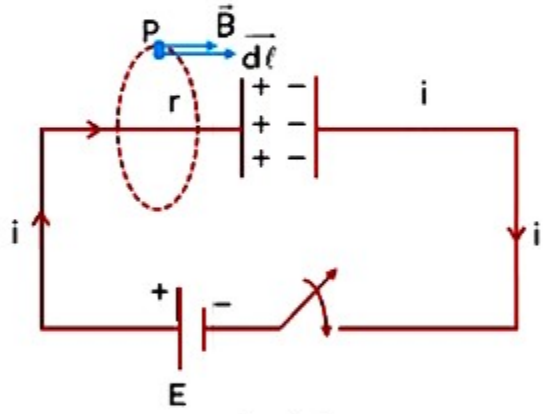


Fig. (a)

According to ampere's Circuits law :-

$$\oint \vec{B} \cdot \overline{d\ell} = \mu_0 I_{in}$$

$$\oint |\vec{B}| |\overline{d\ell}| \cos\theta^\circ = \mu_0 i$$

Since the magnitude of magnetic field (B) is constant at each and every point on amperian loops so we can take B field outside the integral sign.

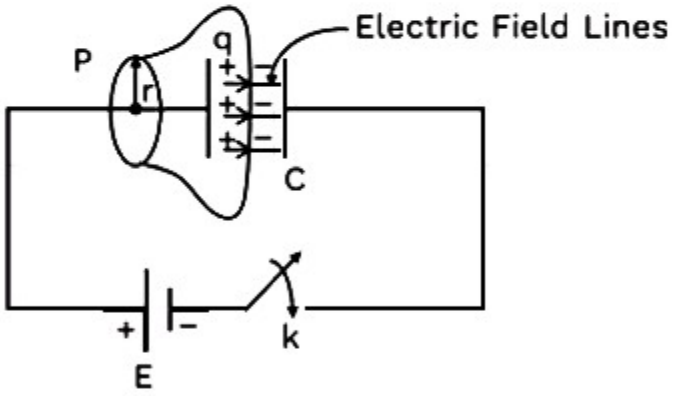
$$|\vec{B}| \oint \overline{d\ell} = \mu_0 i$$

$$|\vec{B}| \times 2\pi r = \mu_0 i$$

The Megnetic field at point P

$$|\vec{B}_P| = \frac{\mu_0}{2\pi} \frac{i}{r}$$

Case-2 : Now consider a different surface (in this case shape of the Amperian loop has changed, shown in the figure) that has the same boundary. This is a pot-like surface.



(fig. b)

Now Acc to Ampere's Law:-

$$\oint \vec{B} \cdot \overline{d\ell} = \mu_0 I_{in}$$

$$\oint |\vec{B}| |\overline{d\ell}| \cos\theta^\circ = \mu_0 (I_{in} = 0)$$

$I_{in} = 0$; Since no current passing through the surface.

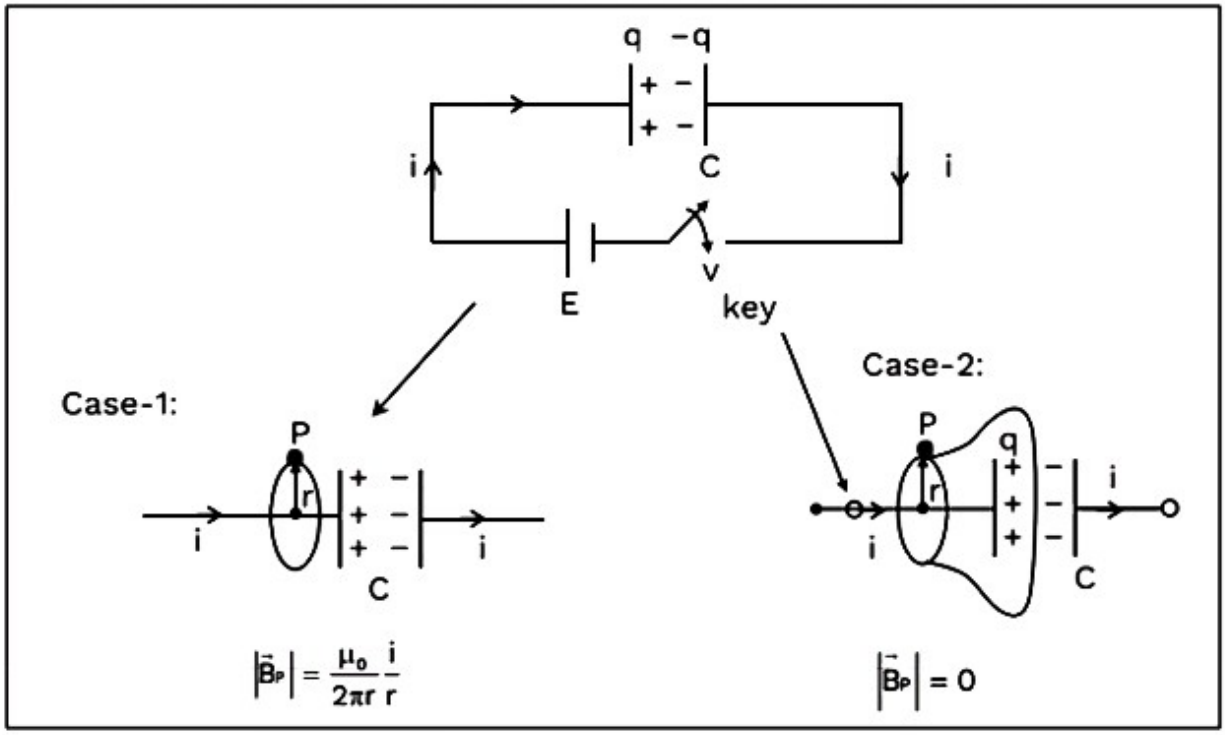
$$|\vec{B}| \oint |\overline{d\ell}| \times 1 = 0$$

Magnetic field at point 'P'.

$$|\vec{B}_P| = 0$$

So, we have a contradiction, calculated one way, there is a magnetic field at point 'P', calculated another way the magnetic field at P is zero. Since $\vec{B}$ - field does not depends on shape of amperial loop but in above cases $\vec{B}$ - field are different which shows inconsisistency in Ampere's Circuital Law.

Summary :-



Conculusion :-

Since - field does not depends on shape of amperian loop but in above cases - field are different which shows **inconsistency in Ampere's Circuital Law**.

Modification By Maxwell in Ampere's Circuital Law :- We can actually guess the missing term by looking carefully at fig-b. Is there anything passing through the surface between the plates of the capacitors; yes of course, the electric field.

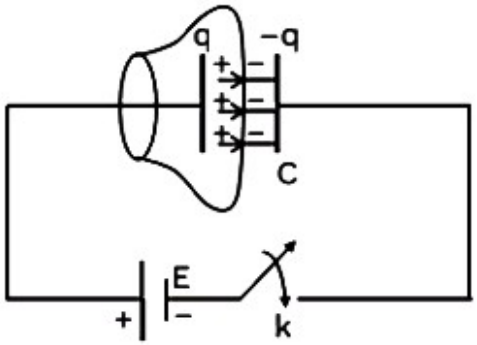


Fig. (b)

Electric field between the plates of capacitors :-

$$\vec{E} = \frac{\sigma}{\epsilon_0} = \frac{q}{A\,\epsilon_0} \qquad ; \text{(charge density } \sigma = \frac{q}{A}\text{)}$$

$$\text{Electric Flux} = \vec{E} \cdot \vec{A} \Rightarrow \phi_E = \left(\frac{q}{A\,\epsilon_0}\right) \times A = \frac{q}{\epsilon_0}$$

$$\phi_E = \frac{q}{\epsilon_0} \qquad \ldots(1)$$

Now if the charge q on the capacitor plates charges with time, there is a current $i = \left(\frac{dq}{dt}\right)$ from equation 1.

$$\frac{d\phi_E}{dt} = \frac{d}{dt}\left(\frac{q}{\epsilon_0}\right)$$

$$\frac{d\phi_E}{dt} = \frac{1}{\epsilon_0} \times \frac{dq}{dt}$$

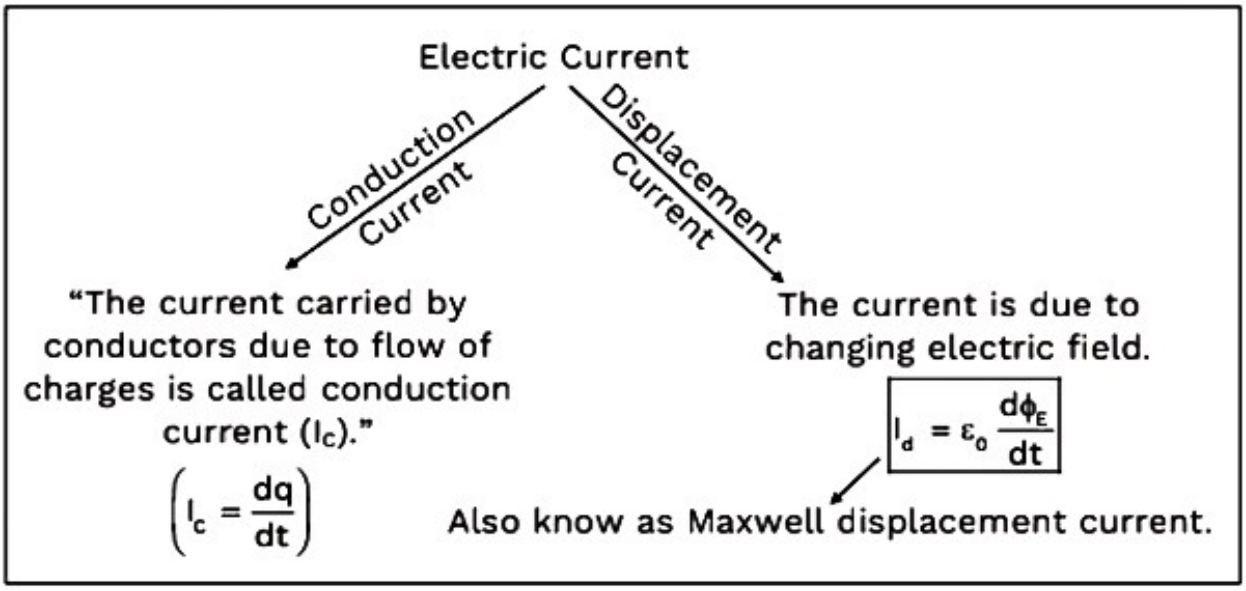
$$\frac{dq}{dt} = \epsilon_0 \frac{d\phi_E}{dt} \qquad ...(2)$$

This implies that for consistence

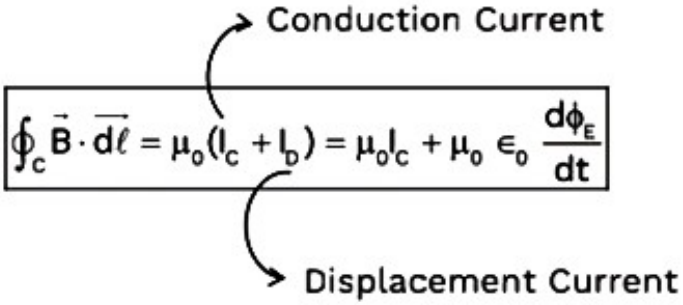
$$\frac{dq}{dt} = i_D = \epsilon_0 \times \frac{d\phi_E}{dt}$$

"This is missing term in Ampere's Circuital Law"

Summary :-

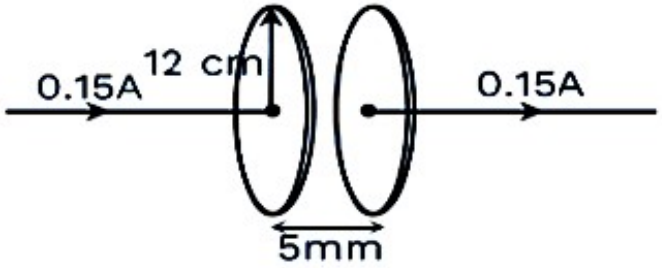


This idea led maxwell to modify Ampere's Circuit Law in order to make the same Logically consistent.



It is now called as **Ampere Maxwell Law**.

Q. **In the given figure, a capacitor made of circular plates having a radius of 12 cm and plate separation of 5 mm is being charged by an external source (not shown in figure). The charging current is 0.15A which is constant in nature.**



(A) **Find out the capacitance and the rate of change of the potential difference between the plates.**

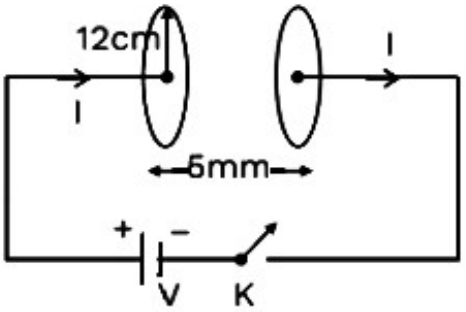
(B) **Obtain the displacement current across the plates.**

(C) **Is Kirchhoff's first rule (Junction Rule) valid at each plates of the capacitor? Explain.**

Sol. (A) Capacitance of the Parallel Plate Capacitor

$$C = \frac{\epsilon_0 A}{A} = \frac{8.85 \times 10^{-12} \times \pi \times (12 \times 10^{-2})^2}{5 \times 10^{-3}}$$

$C = 80.1 \times 10^{-12} F = 80.1$ pF



We know Q = CV

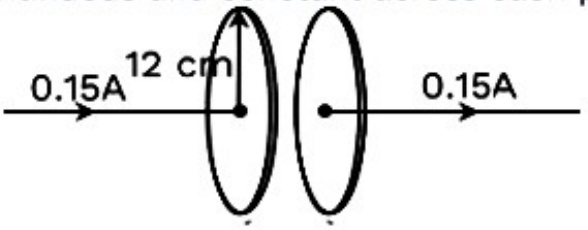
$$I = \frac{dQ}{dt} = C\frac{dV}{dt}$$

So, $\frac{dv}{dt} = \frac{I}{C} = \frac{0.15}{80.1 \times 10^{-12}} = 1.875 \times 10^9$ Volt/Sec

(B) The magnitude of displacement current is equal to conduction current.

∴ Displacement current (I_d) = Conduction Current (I) = 0.15 A

(C) Yes, Kirchhoff's first Law is very much applicable to each plate of capacitor as $I_d = I$.
So current is continuous and constant across each plate.



Q. **Let the electric field amplitude of an electromagnetic wave is $E_0 = 120\ NC^{-1}$ and that its frequency is f = 50.0 MHz.**
(A) Determine, B_0, ω, k, and λ.
(B) Find expression for $\vec{E}$ and $\vec{B}$.

Sol. (A) By relation, $\frac{E_0}{B_0} = C$

Peak value of magnetic field

$$B_0 = \frac{E_0}{C} = \frac{120}{3 \times 10^8} = 40 \times 10^{-8}\ \text{Tesla}$$

Angular frequency, $\omega = 2\pi f$

$$= 100\pi \times 10^6 \text{Hz}.$$

Speed of light $C = f \times \lambda$

Wave length, $\lambda = \frac{C}{f} = \frac{3 \times 10^8}{50 \times 10^6}$; $\boxed{\lambda = 6\text{m}}$

Propagation constant,

$$k = \frac{2\pi}{\lambda} = \frac{\pi}{3} m^{-1}$$

$\boxed{k = 1.05\ m^{-1}}$

(B) Let the wave is propagating along x-direction, electric field $\vec{E}$ is along y-direction and a magnetic field $\vec{B}$ along z-direction.

$$\vec{E}_y = E_0 \sin(kx - \omega t)\hat{j}$$

$$\vec{E}_y = 120 \sin\left(\frac{\pi}{3}x - 100\pi \times 10^6 t\right)\hat{j}$$

$$\vec{B}_z = B_0 \sin(kx - \omega t)\hat{k}$$

$$\boxed{\vec{B}_z = 40 \times 10^{-8} \sin\left(\frac{\pi}{3}x - \pi \times 10^8 t\right)\hat{k}}$$

HERTZ EXPERIMENT (Practical Production of EM Waves)

In 1888, Hertz demonstrated the production of electromagnetic waves in the Lab. His experimental apparatus is shown in the figure.

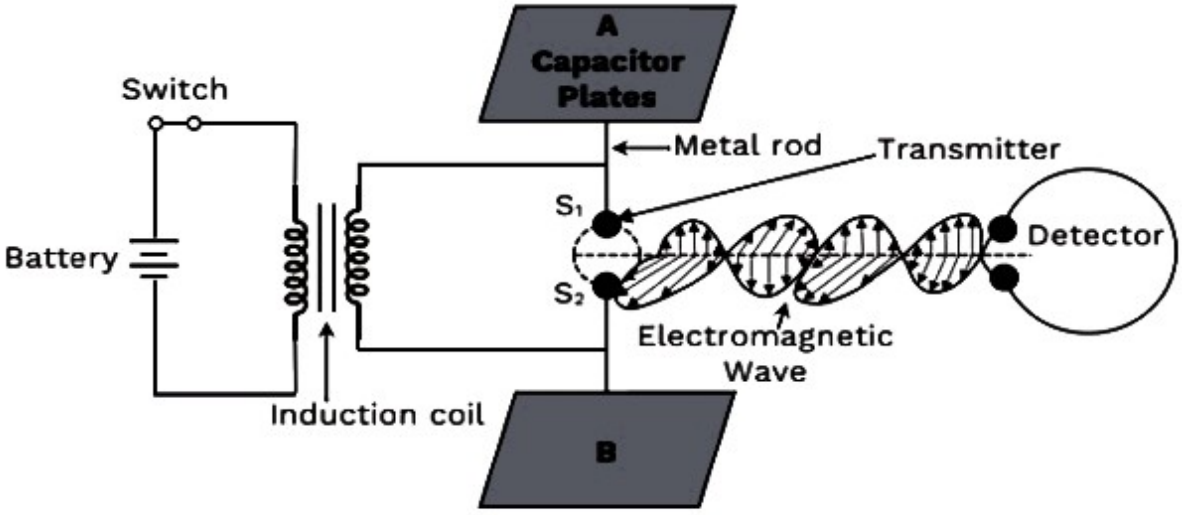


Fig: The apparatus used by Hertz for producing and detecting the electromagnetic waves

Important Point about Hertz Experiments :-

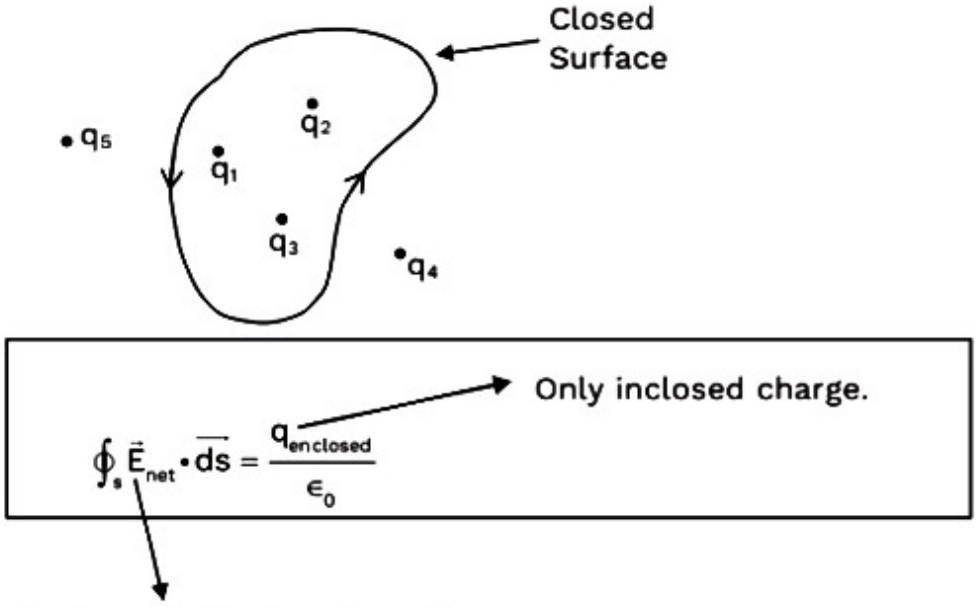
1. Hertz's experiment was based on the fact that an oscillating electric charge radiates electromagnetic waves.
2. The Hertz experiment setup contains two metallic plates of equal size held parallel to each other. These plates are connected to two metal spheres, which act as a transmitter.
3. These spheres are charged to high-value potential with the help of an induction coil.
4. When a high voltage is applied across the metal spheres, they get charged and produce a very high value of an electric field $\left(\vec{E} > 3 \times 10^{6}\, v / m\right)$. Due to this high value of the electric field, breakdown of air occurs, and the air gets ionized, and a high value of electric current starts flowing between spheres, and due to this, sparking is observed.
5. This oscillating charge (sparking) produces electromagnetic waves.
6. Hertz was able to defect these waves using a single loop of wire which acted as a receiver. The EM waves propagated from the transmitter and produced a spark between two open ends of the receiver coil. The spark was a spark between two open ends of the receiver coil. The spark was due to oscillating electric & magnetic fields (EM waves) in the receiver coil.

Maxwell Equations:-

All the basic Laws of electromagnetism can be formulated in four fundamental equations corresponding to four basic Law which are called "Maxwell – Equation".

(1) Maxwell's First Equation : Gauss's Law in Electrostatics:-

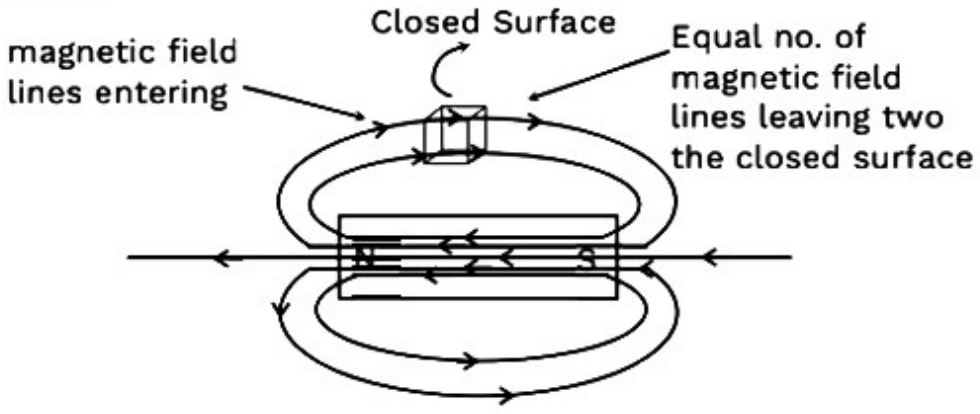
Total electric flux through any closed surface is equal to $\frac{1}{\epsilon_0}$ times the net in closed charge.



(2) Maxwell's Second Equation (Gauss's Law in Magnetism):-

The law states taht the magnetic flux across any closed surface is euql to zero.

$$\oint_S \vec{B} \cdot \overline{ds} = 0$$



Magnetic Field Lines Generating Through Bar Magnet.

No Mono Pole Exists :-

As a magnetic monopole does not exist, always magnetic flux through a closed surface is equal to zero because magnetic field lines are always closed loop lines.

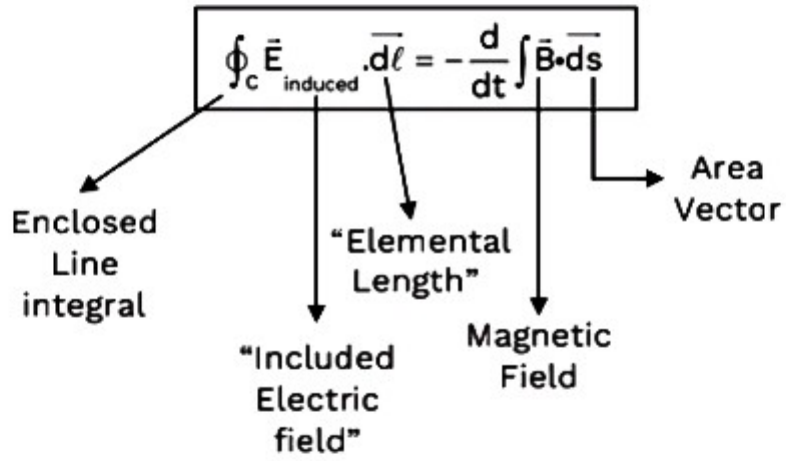
(3) Maxwell's Third Equation (Faraday's Law of Electromagnetic Induction):-

It states that the induced emf produced in a circuit is numerically equal to time rate of charge of magnetic flux through it.

$$\text{Induced emf} = -\frac{d}{dt}(\phi_B)$$

$$\text{Magnetic flux } (\phi_B) = \int \vec{B}\cdot\overrightarrow{ds}$$

$$\text{Induced Emf} = -\frac{d}{dt}\int \vec{B}\cdot\overrightarrow{ds}$$



Maxwell's Fourth Equation (Maxwell-Ampere's Circuital Law):-

Conduction current I and the displacement current I_D together passes the property of continuity along any closed path. Hence

$$\oint \vec{B}\cdot\overrightarrow{d\ell} = \mu_0(I_C + I_D)$$

$$\oint \vec{B}\cdot\overrightarrow{d\ell} = \mu_0\left(I_C + \epsilon_0 \frac{d\phi_E}{dt}\right)$$

→ This is Ampere – Maxwell Law.

Where ;

I_C = Conduction current.

I_D = Displacement Current

Important Point :-

Like conduction current displacement current is also a source of magnetic field.

Even though $(I_C + I_D)$ is always continuous along any closed path, individually I_C and I_D may not be continuous.

Summary :-

(1) Gauss's Law of Electrostatics

$$\oint \vec{E} \cdot \overrightarrow{dA} = \frac{q_{in}}{\epsilon_0}$$

(2) Gauss's Law for Magnetism

$$\oint \vec{B} \cdot \overrightarrow{dA} = 0$$

(3) Faraday's Law

$$\oint \vec{E} \cdot \overrightarrow{d\ell} = -\frac{d\phi_B}{dt} = -\frac{d}{dt}\left(\int \vec{B} \cdot \overrightarrow{dA}\right)$$

(4) Ampere's Law

$$\oint \vec{B} \cdot \overrightarrow{d\ell} = \mu_0 I_c + \mu_0 \epsilon_0 \left(\frac{d\Phi_E}{dt}\right)$$

$$\oint \vec{B} \cdot \overrightarrow{d\ell} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d}{dt}\int \vec{E} \cdot \overrightarrow{dA}$$

Maxwell Equation In Free Space :-

$I_c = 0 \rightarrow$ No conduction current in free space.

$q_{in} = 0$ (No charge in free space)

1. $\oint \vec{E} \cdot \overrightarrow{dA} = 0$
2. $\oint \vec{B} \cdot \overrightarrow{dA} = 0$
3. $\oint \vec{E}_{ind} \cdot \overrightarrow{d\ell} = -\frac{d}{dt}\int \vec{B} \cdot \overrightarrow{dA}$
4. $\oint \vec{B} \cdot \overrightarrow{d\ell} = \mu_0 \epsilon_0 \frac{d}{dt}\int \vec{E} \cdot \overrightarrow{dA}$

Q. **Choose the correct option**

Which conclusion we can draw on the basis of maxwell's hypothesis "a changing electric field gives rise to"

(A) an emf **(B) Electric Current**

(C) Magnetic Field **(D) Pressure Radiant**

Sol. **Ans.** (C)

According to maxwell's law time-varying electric field produces a magnetic field, and the time-varying magnetic field produces an electric field.

Q. **Maxwell's equation $\oint \vec{B} \cdot \overline{d\ell} = -\frac{d\phi_B}{dt}$ is a statement of.**

(A) Ampere's Law
(B) Faraday's Law of Induction
(C) Gauss's Law of Electricity
(D) Gauss's Law of Magnetism

Sol. **Ans.** (B)
According to Faraday's law of electromagnetic induction Negative rate of change of magnetic flux is equal to induced emf.

Equation of Electromagnetic Wave Using Maxwell Equation:-

For Free Space : No charge (q) = 0

No current $i_c = 0$

Now can write maxwell's equations –

(1) $\oint \vec{E} \cdot \overline{dA} = 0$

(2) $\oint \vec{B} \cdot \overline{dA} = 0$

(3) $\oint \vec{E}_{ind} \cdot \overline{d\ell} = -\frac{\partial}{\partial t} \int \vec{B} \cdot \overline{dA}$

(4) $\oint \vec{B} \cdot \overline{d\ell} = \mu_0 \epsilon_0 \frac{\partial}{\partial t} \int \vec{B} \cdot \overline{dA}$

The electromagnetic wave equation is a second-order partial differential equation that describes the propagation of Electromagnetic wave through a medium or in a vacuum.

The homogeneous form of the equation, written in terms of either the electric field $\vec{E}$ or the magnetic field $\vec{B}$.

The wave equation can be derive using above maxwell's equation.

Using the maxwell equation, we can prove that an EM wave travelling along x-axis having an electric field varying sinusoidally along the y-axis will give rise to magnetic field varying sinusoidally along z-axis.

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2}$$

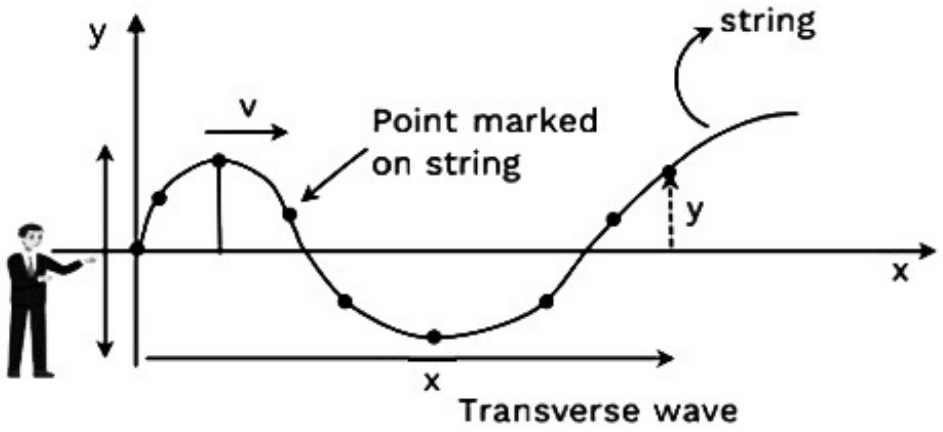
$$\frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2}$$

$$\frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2} \quad ...(1)$$

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} \quad ...(2)$$

Second order partial differential equation of EM wave.

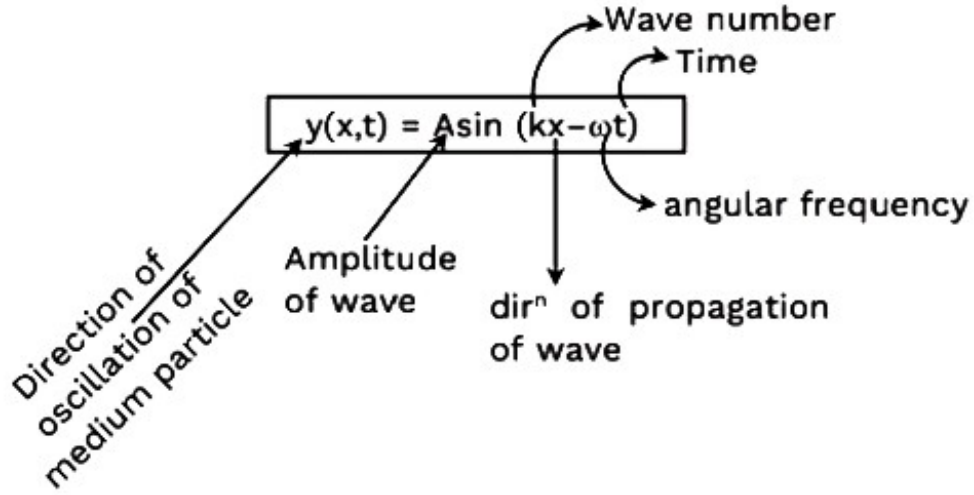
General Equation of Wave:-



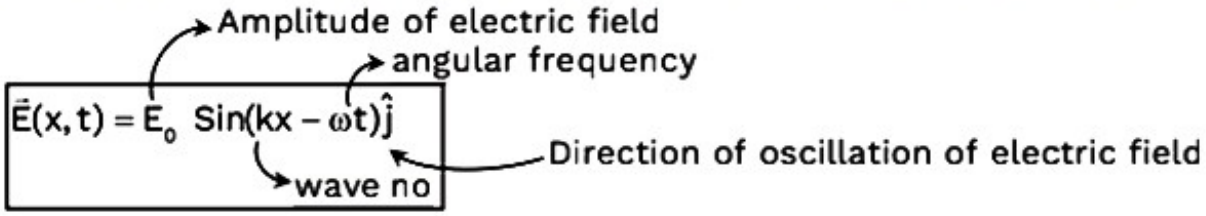
Differential equation form of transverse wave

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \Rightarrow \boxed{\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}} \qquad ...(3)$$

We can write the Solution of this differential equation:-



Similarly, the Solution of equation (1) and equation (2) can be written as :-



$$\vec{B}(x,t) = B_0 \sin(kx - \omega t)\hat{k}$$

"direction of oscillation of magnetic field."

where ; $k = \frac{2\pi}{\lambda}$ → Wave length

$\omega = 2\pi f$ → frequency of EM wave

angular frequency

$v = f \times \lambda = \frac{\omega}{k}$

Speed of EM wave

→ wave no

Now compare equation (1) and equation (3)

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Speed of electromagnetic wave in vacuum also denoted by 'c'.

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

$$\frac{1}{4\pi \epsilon_0} = 9 \times 10^9$$

$$\epsilon_0 = \frac{1}{4\pi \times 9 \times 10^9}$$

Put the value of μ_0 and ϵ_0

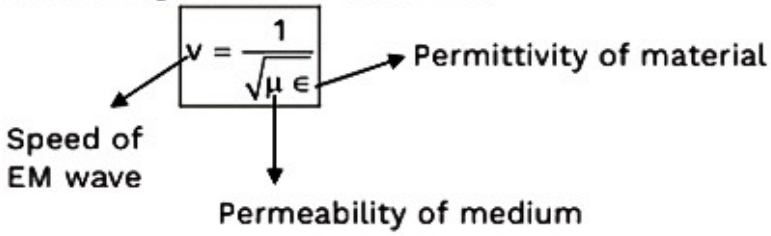
$$C = \frac{1}{\sqrt{4\pi \times 10^{-7} \times \frac{1}{4\pi \times 9 \times 10^9}}}$$

$$= \frac{\sqrt{9 \times 10^9}}{\sqrt{10^{-7}}} = \sqrt{9 \times 10^{16}}$$

$C = 3 \times 10^8$ m / sec Speed of EM wave in vacuum.

Results of Maxwell's Equation:-

(1) In a material medium of permittivity ϵ and magnetic permeability 'μ', velocity of electromagnetic wave becomes;



From relation ; $\mu = \mu_0\mu_r$; $\epsilon = \epsilon_0\epsilon_r$

$$v = \frac{1}{\sqrt{\mu_0\mu_r \cdot \epsilon_0\epsilon_r}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \times \frac{1}{\sqrt{\mu_r \epsilon_r}}$$

$$\boxed{v = \frac{C}{\sqrt{\mu_r \epsilon_r}}}$$

Where,

$C = 3 \times 10^8$ m/sec = Speed of EM wave in free space

v = Speed of EM wave in medium

μ_r = Relative Permeability of medium.

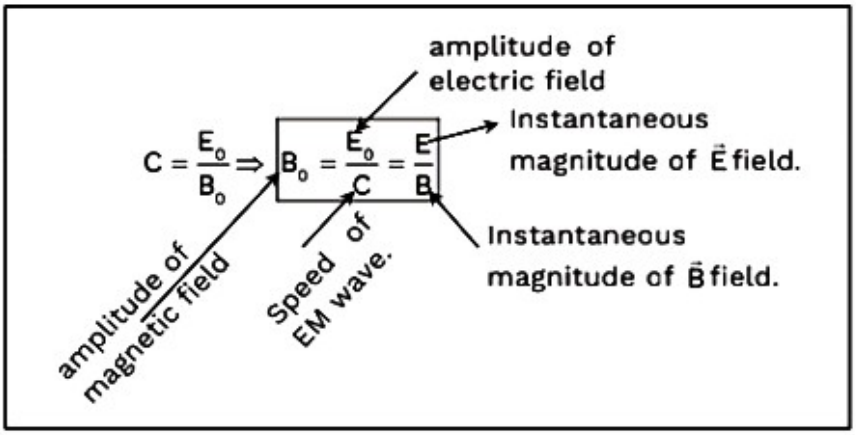
ϵ_r = Relative Permittivity of medium.

So we can conclude that the velocity of light depends on the electric & magnetic properties of the medium.

(2) The instantaneous magnitudes of $\vec{E}$ & $\vec{B}$ in on electromagnetic wave are related by expression.

$$\boxed{\frac{E}{B} = C}$$

Also relation $\frac{E_0 \sin(\omega t - kx)}{B_0 \sin(\omega t - kx)} = C$



Properties of Electromagnetic Wave and Important Relations:-

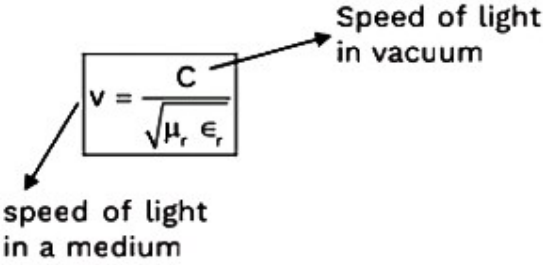
1. It is transverse wave.
2. It is propagate in free space.
3. A medium can change its propagation properties.
4. When E.M.W. travelled in space there is variation of electric and magnetic field in space.
5. Electromagnetic wave carry energy and momentum.
6. Electric and magnetic fields are perpendicular to the direction of propagation of the electromagnetic wave.
7. An electric field is perpendicular to .
8. The direction $\vec{E} \times \vec{B}$ is along the direction of propagation of the electromagnetic wave.
9. Electric field and magnetic field are in phase.
10. In electromagnetic waves, the electric field vector $\vec{E}$, magnetic field vector $\vec{B}$, and propagation vector $\vec{p}$ are mutually perpendicular such that $\vec{E}$, $\vec{B}$ and $\vec{p}$ from a right angle system. Hence electromagnetic waves are transverse in nature.
11. Electromagnetic waves travel with speed of light. In a vacuum, their speed is

$$\boxed{C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}}$$

In isotropic medium, their speed is

$$v = \frac{1}{\sqrt{\mu \epsilon}} = \frac{1}{\sqrt{\mu_r \mu_0 \epsilon_0 \epsilon_r}}$$

$$= \frac{1}{\sqrt{\mu_r \epsilon_r}} \cdot \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{C}{\sqrt{\mu_r \epsilon_r}}$$



12. The medium offers hindrance to the propagation of wave. Such a hindrance is called wave impedance (z), and its value in a medium is given by

$$z = \frac{E}{H} = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0 \mu_r}{\epsilon_0 \epsilon_r}}$$

$$\boxed{z = \sqrt{\frac{\mu_r}{\epsilon_r}} \times \sqrt{\frac{\mu_0}{\epsilon_0}}}$$

"Intrinsic Impedance of the medium"

$\Rightarrow$ The intrinsic impedance of free space

$$z = \sqrt{\frac{\mu_0}{\epsilon_0}} = 377\,\Omega \quad \Rightarrow \quad \boxed{Z = 377\sqrt{\frac{\mu_r}{\epsilon_r}}\,\Omega}$$

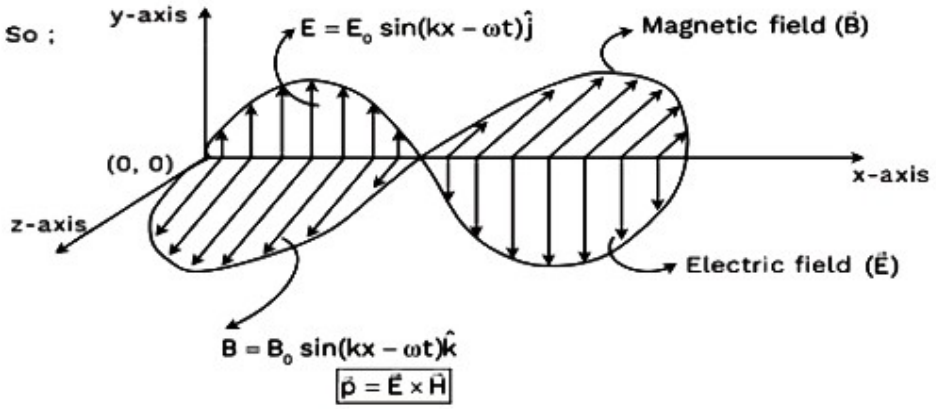
13. The electric and magnetic fields of a sinusoidal plane electromagnetic wave propagation in the positive – x- direction can also be written as –

$\vec{E} = E_0 \sin(kx - \omega t)\hat{j}$

$\vec{B} = B_0 \sin(kx - \omega t)\hat{k}$

Where ω is the angular frequency and k is the angular wavenumber or propagation constant, which are given by;

$$\boxed{\omega = 2\pi f \;;\; k = \frac{2\pi}{\lambda}}$$



$$\vec{p} = \left[E_0 \sin(kx - \omega t)\hat{j}\right] \times \left[H_0 \sin(kx - \omega t)\hat{k}\right]$$

Poynting Vector (It shows the intensity of wave and direction of E-M wave propagation)

direction of oscillation of electric field

direction of oscillation of magnetic field

14. over one cycle, the average value of the Poynting vector taken is equal to the the intensity of a sinusoidal plane electromagnetic wave.

Q. **The oscillating electric field was found at 18 V/m. Find the magnitude of the oscillating magnetic field will be**

(A) 4×10^{-6}T

(B) 6×10^{-8}T

(C) 9×10^{-9}T

(D) 11×10^{-11}T

Sol. **Ans.** (B)

$$C = \frac{E}{B} \Rightarrow B = \frac{E}{C} = \frac{18}{3 \times 10^8} = 6 \times 10^{-8} T$$

Q. **The electric and magnetizing fields of an electromagnetic are 10 Am^{-1} and 0.265 Am^{-1}. Find the maximum energy flow.**

(A) 36.5 W/m^2

(B) 26.5 W/m^2

(C) 46.7 W/m^2

(D) 765 W/m^2

Sol. **Ans.** (B)

Given that $E_0 = 100$ V/m , $B_0 = 0.265$ A/m.

$\therefore$ Maximum energy flow rate can be calculated as

$$S = E_0 \times B_0$$
$$= 100 \times 0.265$$
$$= 26.5 \text{ w/m}^2$$

Q. **In electromagnetic waves, the oscillating magnetic and electric field vectors are oriented along.**

(A) The same direction and are in phase.

(B) The same direction but differ in phase by 90°.

(C) Mutually perpendicular directions and differ in phase by 90°.

(D) Mutually perpendicular directions and are in phase.

Sol. **Ans.** (D)

$\vec{E}$ and $\vec{B}$ are mutually perpendicular to each other and are in phase, i.e. and they become and minimum at the same place and at the same time.

Q. **Let an electromagnetic wave travels along z-axis. Which of the following pairs of space and time-varying fields would generate such a wave**
(A) E_x, B_y (B) E_y, B_x (C) E_z, B_x (D) E_y, B_y

Sol. **Ans.** (A)
E_x and By would generate a plane Electromagnetic wave travelling in z-direction.
$\vec{E}$, $\vec{B}$ and $\vec{S}$ form a right-handed system $\vec{K}$ is along z-axis. As $\hat{i} \times \hat{j} = \hat{k}$

$\Rightarrow Ex\hat{i} \times By\hat{k} = C\hat{k}$ ie. E is along x-axis, and B is along y-axis.

Q. **Suppose a plane electromagnetic wave is incident on a material surface. It the wave delivers momentum p and energy E, then**
(A) $p = 0, E = 0$ (B) $p \neq 0, E \neq 0$ (C) $p \neq 0, E = 0$ (D) $p = 0, E \neq 0$

Sol. **Ans.** (B)
Electromagnetic waves carry momentum and hence can exert pressure on surfaces. They also transfer energy to the surface so $p \neq 0$ and $E \neq 0$.

Energy In Electromagnetic Wave:-

Electrostatic Energy Density :

$$\boxed{U_E = \frac{1}{2}\epsilon_0 E^2} \qquad ...(1)$$

Magnetic Energy Density :

$$\boxed{U_B = \frac{B^2}{2\mu_0}} \qquad ...(2)$$

From Relation : $kE_0 = \omega B_0$

$$B_0 = \frac{k}{\omega} \times E_0$$

$$\boxed{C = \frac{\omega}{k} = \text{Velocity of electromagnetic wave of vacume}}$$

$$\boxed{B_0 = \frac{E_0}{C}} \qquad ...(3)$$

From Magnetic Energy Density Expression:-

$$\boxed{U_B = \frac{1}{2\mu_0}(B^2)}$$

$$U_B = \frac{1}{2\mu_0} \times B_0^2 \sin^2(\omega t - kx) \quad \left\{\begin{matrix}\text{Instantaneous Value of magnetic} \\ \text{field } B = B_0 \sin(\omega t - kx)\end{matrix}\right\}$$

$$U_B = \frac{1}{2\mu_0} \times \left(\frac{E_0}{C}\right)^2 \sin^2(\omega t - kx)$$

$$U_B = \frac{1}{2\mu_0} \times \left(\frac{1}{C^2}\right) \times E_0^2 \times \sin^2(\omega t - kx)$$

$$U_B = \frac{1}{2\mu_0} \times \mu_0 \epsilon_0 \times E_0^2 \times \sin^2(\omega t - kx)$$

$$\boxed{U_B = \frac{1}{2}\epsilon_0 \times E_0^2 \sin(\omega t - kx)}$$

(E_0 → Amplitude of Electric field)

$$\boxed{U_B = \left(\frac{1}{2} \times \epsilon_0 E^2\right)}$$

(E → Instantaneous Electric Field)

Conclusion :-

On the basis of previous derivation, both energy density are equal:-

$\boxed{U_B = U_E}$

U_E → Electric Energy Density

U_B → Magnetic Energy Density

Total Energy Density :-

$$U_T = U_B + U_E$$

$$U_{Total} = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\epsilon_0 E^2 \quad ; \quad \left\{\text{Since } U_B = U_E = \frac{1}{2}\epsilon_0 E^2\right\}$$

$\boxed{U_{Tctal} = (\epsilon_0 E^2)}$ → Instantaneous value of the electric field.

$$\boxed{U_{Total} = \epsilon_0 E_0^2 \sin^2(\omega t - kx)}$$

Instantaneous Value of Total Energy Density.

⇒ Maximum Value of Energy Density = $\epsilon_0 E^2$

Time average of Energy Density :-

$$< U_T >= \frac{1}{T}\int_0^T U \cdot dt$$

$$= \frac{1}{T}\int_0^T \epsilon_0 E_0^2 \sin^2(\omega t - kx)$$

$$= \epsilon_0 E_0^2 \left[\frac{\int_0^T \sin^2(\omega t - kx)dt}{T}\right]$$

$$= \epsilon_0 E_0^2 \left(\frac{1}{2}\right)$$

$\boxed{< U_T >= \frac{1}{2}\epsilon_0 E_0^2}$ ← Important Relation

Average value of total energy density

Summary :-

1. Instantaneous value of energy density

$\epsilon_0 E^2$

$U_E = \frac{1}{2}\epsilon_0 E^2$ — Instantaneous value of electric energy density

$U_B = \frac{B^2}{2\mu_0} = \frac{1}{2}\epsilon_0 E^2$ — Instantaneous value of Magnetic energy density

2. Instantaneous value total energy density

$\epsilon_0 E^2$ — Maximum Value of Total Energy Density

$\frac{1}{2}\epsilon E_0^2$ (Maximum Value of Electric Field) Energy density

$\frac{1}{2}\epsilon E_0^2$ (Maximum Value of Magnetic Energy Density)

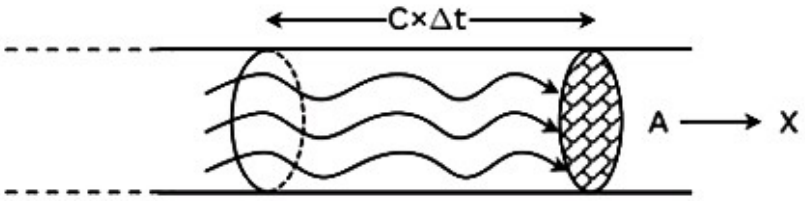
3. Average Energy Density :

$< U_T > = \frac{1}{2}\epsilon_0 E_0^2$

$\frac{1}{4}\epsilon_0 E_0^2$ Average Value of Electric Field Energy Density

$\frac{1}{4}\epsilon_0 E_0^2$ "Average Value of Magnetic Field Energy

Intensity of Electromagnetic wave :- The energy crossing per unit area per unit time perpendicular to direction of propagation is called the intensity of a wave.



Consider a cylindrical volume with area of cross-section A and length CΔt along the x-axis. The energy contained in this cylinder crosses the area A in time Δt as the wave propagates at speed 'c'.

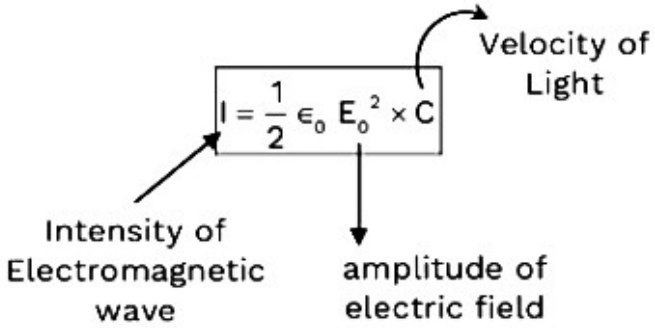
The Energy Contained is

$$U = \langle U_{avg} \rangle \times (C \times \Delta t) \times A$$

The Intensity is

$$I = \frac{U}{A \times \Delta t} = \frac{\langle U \rangle \times C \times \Delta t \times A}{A \times \Delta t}$$

$$\boxed{I = \langle U \times C}$$



In Terms of $\vec{B}$-field:

$$B_0 = \frac{E_0}{C}$$

$$I = \frac{1}{2} \epsilon_0 \times (B_0 \times C)^2 \times C$$

From Relation $\boxed{C^2 = \frac{1}{\mu_0 \epsilon_0}}$

$$I = \frac{1}{2} \epsilon_0 \times B_0^2 \times \frac{1}{\mu_0 \epsilon_0} \times C$$

$$\boxed{I = \left(\frac{B_0^2}{2\mu_0}\right) C}$$

Q. **The electromagnetic waves do not transport.**
(A) energy (B) charge (C) momentum (D) Information

Sol. **Ans.** (B)
EM waves transport energy-momentum and information but not charge. EM waves are unchanged.

Q. **A radio receiver antenna that is 2-meter-long is oriented along the direction of the electromagnetic wave and receives a signal of intensity 5×10^{-16} watt/ m^2. Find the maximum instantaneous potential difference across the two ends of the antenna is**
(A) 1.23 μv **(B) 1.23 mv (C) 1.23 v (D) 12.3 mv.**

Sol. **Ans.** (A)

$$I = \frac{1}{2}\epsilon_0 CE_0^2$$

$$\Rightarrow \quad E_0 = \sqrt{\frac{2I}{\epsilon_0 C}} = \sqrt{\frac{2\times 5\times 10^{-16}}{8.85}} = 0.61\times 10^{-6}\,\frac{V}{m}$$

$$\text{Also, } E_0 = \frac{V_0}{d} \Rightarrow V_0 = E_0 d = 0.61\times 10^{-6}\times 2$$

$$= 1.23\,\mu V.$$

Q. **About 5% of the power of a 100w light bulb is converted to visible radiation. What is the average intensity of visible radiation.**
(A) at a distance of 1m from the bulb?
(B) at a distance of 10m?
Assume that the radiation is emitted isotropically and neglect reflection.

Sol. The bulb as a point source, radiate light in all direction uniformly and it is given that only 5 % of power is converted to visible radiations.

(A) Let us assume a sphere a radius 1m surface area $A = 4\pi r^2 = 4\pi(1)^2 = 4\pi m^2$

Intensity on the Sphere :-

$$I = \frac{\text{Power}}{\text{Area}} = \frac{100\times 0.05}{4\pi} = 0.4\,W/m^2$$

(B) Let us assume a sphere of 10m radius. Surface area $A = 4\pi r^2 = 4\pi(10)^2 = 400\pi m^2$

Intensity on the Sphere :-

$$I = \frac{\text{Power}}{\text{Area}} = \frac{100 \times 0.05}{400\pi} = 0.004\ w/m^2$$

Q. **In a plane electromagnetic wave have an amplitude of $48Vm^{-1}$, and the electric field oscillates sinusoidally at a frequency of 2.0×10^{10} Hz.**

(A) what is the wavelength of the wave?

(B) What is the amplitude of the oscillating magnetic field?

(C) Show that the average energy density of the $\vec{E}$ field equals the average energy density of the $\vec{B}$ field. [$C = 3 \times 10^8\ ms^{-1}$]

Sol. (A) wavelength of electromagnetic wave is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{2 \times 10^{10}} = 1.5 \times 10^{-2} m$$

(B) Amplitude of magnetic field,

$$\frac{E_0}{B_0} = C$$

$$B_0 = \frac{E_0}{C} = \frac{48}{3 \times 10^8} = 16 \times 10^{-8}\ T$$

(C) Energy density as electric field,

$$U_E = \frac{1}{2} \epsilon_0 E^2$$

Here, $\frac{E}{B} = C$

So, $U_E = \frac{1}{2} \epsilon_0 C^2 B^2$

Where speed of electromagnetic wave,

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

So, $U_E = \frac{1}{2} \frac{\epsilon_0}{\mu_0 \epsilon_0} B^2$

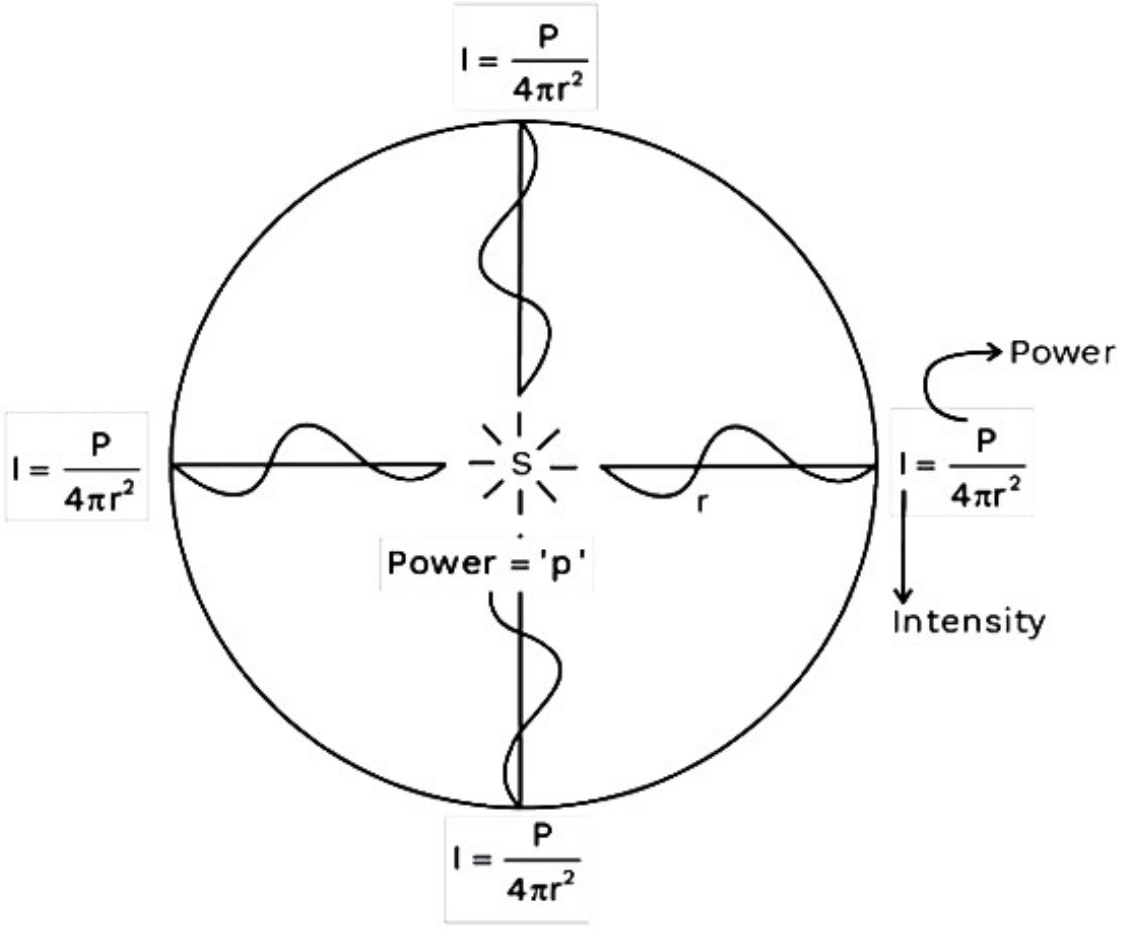
$$U_E = \frac{B^2}{2\mu_0} = U_B$$

$$U_E = \frac{B^2}{2\mu_0} = U_B$$

Electric Energy Density

Magnetic Energy Density

Relation Between Power of a Spherical Source and Intensity :-



Q. **About 2.5% of the power of a 100 watt light bulb is converted to visible radiation. What is the average intensity of the radiation?**

(A) **at a distance of 1m from the bulb.**

(B) **at a distance of 10m from the bulb.**

Assume that radiation is emitted isotropically and neglect reflections.

Sol. Power of visible radiation

$$p' = \frac{2.5}{100} \times 200 = 5w$$

(i) Intensity at distance of 10 m

$$I = \frac{P}{4\pi r^2} = \frac{5}{4 \times 3.14 \times 12} = 0.398 \times 10^{-3} \frac{w}{m^2}$$

(ii) Intensity at 10 m

$$I = \frac{P}{4\pi r^2} = \frac{5}{4 \times 3.14 \times (10)^2} = 0.00398 = 3.98 \times 10^{-3}\ w/m^2.$$

Poynting Vector :-

The rate of energy flow through a unit area perpendicular to the direction of electromagnetic wave propagation is know as Poynting Vector.

If dU energy passes through dA in time interval dt, then Poynting vector can be written as

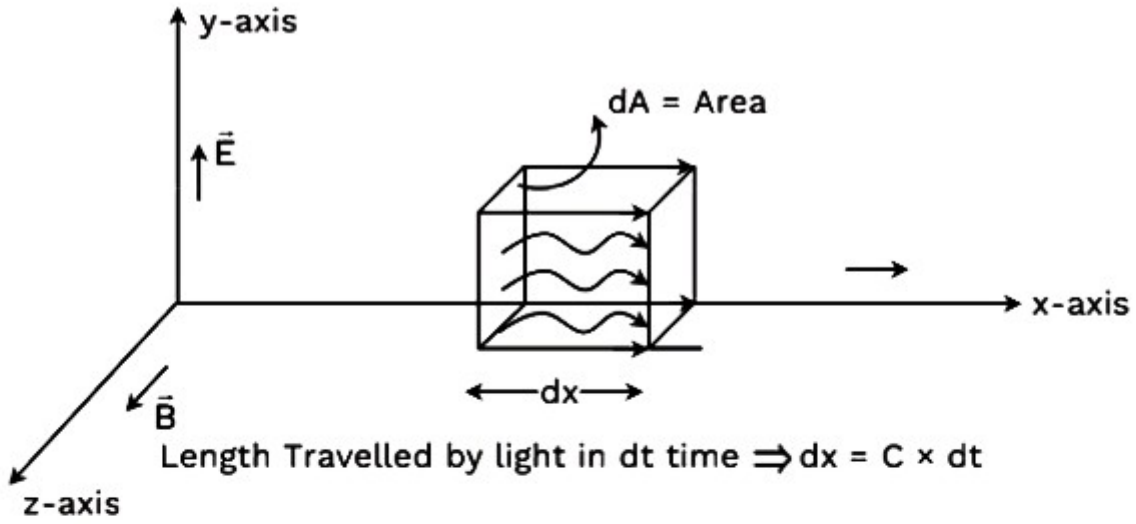
$$\boxed{S = \frac{dU}{dAdt}}; \quad \text{Unit} \frac{J}{m^2 \times Sec} = \frac{watt}{m^2}$$

Vector Relation :-

$$\boxed{\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}}$$

or

$$\boxed{\vec{S} = \vec{E} \times \vec{H}}$$



Length Travelled by light in dt time $\Rightarrow dx = C \times dt$

Volume of infinitely small element

$dV = dA \times dx$

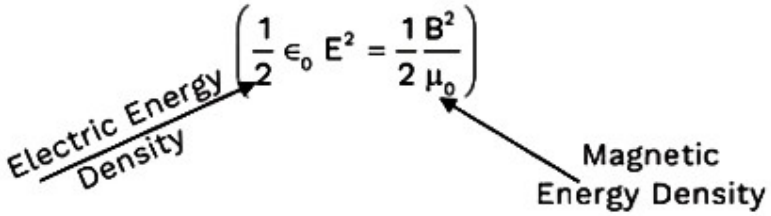
$\Rightarrow$ Total energy contained in the infinitely small Element = Energy density × Volume of Element

Energy = $(U_{Total}) \times dV$

$= (U_E + U_B) \times dV$

$$\text{Energy} = \left(\frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\frac{B^2}{\mu_0}\right) dV$$

Since :-



$$\text{Energy} = \left(\frac{1}{2}\epsilon_0 E^2 + \frac{1}{2}\epsilon_0 E^2\right) dV \qquad \text{Energy} = \left(\epsilon_0 E^2\right) dV$$

$$\text{Energy} = \epsilon_0 E^2 \times dx \times dA \quad \Rightarrow \quad \text{Energy} = \epsilon_0 E^2 \times (C \times dt) \times dA$$

$$\frac{\text{Energy}}{dA \times dt} = \epsilon_0 E^2 \times C \quad \Rightarrow \quad \text{Poynting Vector}\left(|\vec{S}|\right) = \epsilon_0 E^2 C$$

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Poynting Vector → $\boxed{|\vec{S}| = \epsilon_0 E^2 C}$ ...(1)

$$C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \Rightarrow \epsilon_0 = \frac{1}{\mu_0 C^2}$$

$$|\vec{S}| = \frac{1}{\mu_0 C^2} \times E^2 \times C$$

$$\boxed{|\vec{S}| = \frac{E^2}{\mu_0 C}} \qquad \text{...(2)}$$

$$E = BC$$

$$|\vec{S}| = \frac{1}{\mu_0 C^2} \times B^2 C^2 \times C$$

$$\boxed{|\vec{S}| = \left(\frac{B^2}{\mu_0}\right) \times C} \qquad \text{...(3)}$$

Poynting Vector

$$C = \frac{E}{B}$$

$$|\vec{S}| = \frac{E^2}{\mu_0 \times \frac{E}{B}} \qquad \text{(from equation 2)}$$

$$|\vec{S}| = \frac{EB}{\mu_0}$$

$$\boxed{\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}} \qquad ...(4)$$

or

$$\boxed{\vec{S} = \vec{E} \times \vec{H}} \qquad ...(5)$$

Average Value of Poynting Vector :-

From relation ; $\boxed{\vec{S} = \vec{E} \times \vec{H}}$

or

Poynting Vector $\boxed{\vec{S} = \frac{\vec{E} \times \vec{B}}{\mu_0}}$

$$|\vec{S}| = \frac{E_0 \sin(\omega t - kx) \times B_0 \sin(\omega t - kx)}{\mu_0}$$

$$|\vec{S}| = \frac{(E_0 \times B_0) \sin^2(\omega t - kx)}{\mu_0}$$

Average Poynting Vector :-

$$< |\vec{S}| > = \frac{E_0 \times B_0}{\mu_0} < \sin^2(\omega t - kx) >$$

$$< |S| > = \frac{1}{2} \times \frac{E_0 B_0}{\mu_0} \qquad \left\{ \begin{matrix} \text{Since} \\ < \sin^2 \theta > = \frac{1}{2} \end{matrix} \right\}$$

$$\boxed{< |\vec{S}| > = \frac{E_0 B_0}{2\mu_0}}$$

$$\boxed{\text{Intensity} = < |\vec{S}| > = \frac{E_0 B_0}{2\mu_0}} \qquad ...(1)$$

$$< |\vec{S}| > = \frac{E_0 B_0}{2\mu_0}$$

$$B_0 = \frac{E_0}{C}; \qquad C = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \Rightarrow \mu_0 = \frac{1}{C^2 \times \epsilon_0}$$

$$< |\vec{S}| > = \left\{ \frac{E_0 B_0}{2 \times \frac{1}{C^2 \epsilon_0}} \right\}$$

$$= \frac{1}{2} \times C^2 \times \epsilon_0 \times E_0 B_0$$

$$B = \frac{E_0}{C}$$

From relation

$$< |\vec{S}| >= \frac{1}{2} C^2 \times \epsilon_0 \times E_0 \times \frac{E_0}{C} \quad ...(2)$$

The Intensity of EMW Wave =	$< \|\vec{S}\| >= \frac{1}{2} C \epsilon_0 E_0^2$

Intensity = Average Value of Poynting Vector

$$= \frac{1}{2} C \epsilon_0 E_0^2$$

Since ; $B_0 = \frac{E_0}{C} \Rightarrow E_0 = CB_0$

$$C^2 = \frac{1}{\mu_0 \epsilon_0} \Rightarrow \epsilon_0 = \frac{1}{C^2 \mu_0}$$

Intensity $= < |\vec{S}| >= \frac{1}{2} \times C \times \frac{1}{C^2 \times \mu_0} \times (CB_0)^2$

$$= \frac{1}{2} \times C \times \frac{1}{C^2 \times \mu_0} \times C^2 \times B_0^2$$

$$\text{Intensity} = \left(\frac{B_0^2}{2\mu_0} \right) \times C$$

Intensity = Average Value of Poynting Value

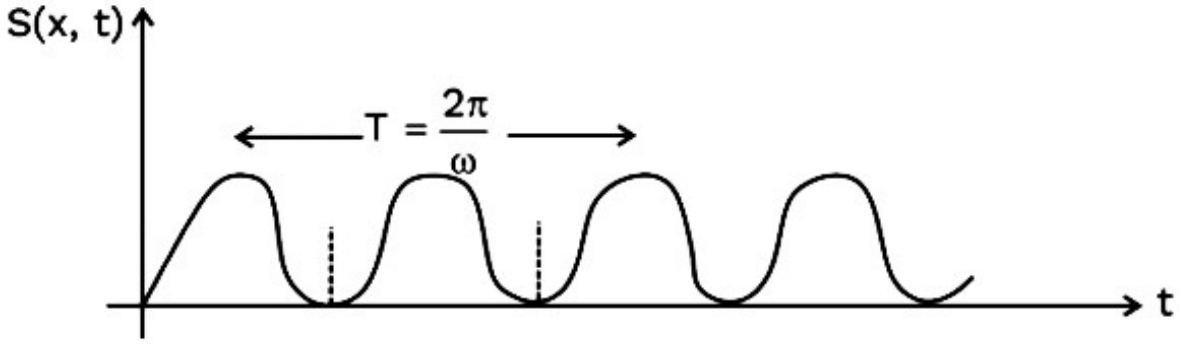
$$=< |\vec{S}| >= \left(\frac{1}{2} \epsilon_0 E_0^2 \right) \times C = \left(\frac{B_0^2}{2\mu_0} \right) \times C$$

Q. **Poynting vector $\vec{S}$ is defined as a vector whose magnitude is equal to the wave intensity and whose direction is along the direction of wave probation. Mathematically, It is given by $\vec{S} = \left(\frac{\vec{E} \times \vec{B}}{\mu_0} \right)$. Show the Nature of $\vec{S}$ Vs t graph.**

Sol. $\vec{E} = E_0 \sin(\omega t - kx)\hat{j} \quad \vec{B} = B_0 \sin(\omega t - kx)\hat{k}$

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) = \frac{1}{\mu_0} E_0 B_0 \sin^2(\omega t - kx)(\hat{j} \times \hat{k})$$

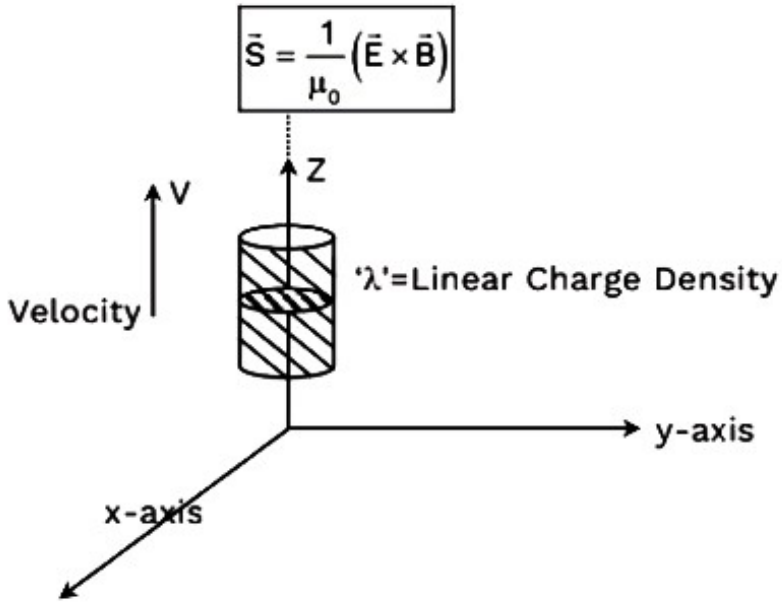
$$\vec{S} = \frac{E_0 B_0}{\mu_0} \sin^2(\omega t - kx)\hat{i}$$



Q. **An infinitely long thin wire carrying a uniform linear static charge density λ is placed along the z-axis (shown in figure). The wire is set into motion along its length with a uniform velocity $\vec{V} = v\hat{k}$. Calculate the Poynting vector.**

Sol. Electric field produced due to infinitely long charged wire,

$$\vec{E} = \frac{\lambda}{2\pi \epsilon_0 a}\hat{j} \quad ...(1)$$



a = radius of cylindrical Gaussian surface around wire.
Magnetic field at .a. distance from current carrying conductor,

$$\vec{B} = \frac{\mu_0 I}{2\pi a}\hat{i} \qquad \text{But}$$

$$I = \frac{q}{t} = \frac{\lambda L}{t} = \lambda v \left[\because Q = \lambda L \text{ and } \frac{L}{t} = v\right]$$

Here L = length

$$\therefore \vec{B} = \frac{\mu_0 \lambda v}{2\pi a} \qquad ...(2)$$

Now pointing vector,

$$S = \frac{1}{\mu_0}\left(\vec{E} \times \vec{B}\right)$$

$$\therefore S = \frac{1}{\mu_0}\left[\frac{\lambda}{2\pi a}\hat{j} \times \frac{\mu_0 \lambda v}{2\pi a}\hat{i}\right]$$

$$= \frac{1}{\mu_0}\left(\frac{\lambda}{2\pi a} \times \frac{\mu_0 \lambda v}{2\pi a}\right)\left(\hat{j} \times \hat{i}\right)$$

$$= \frac{\lambda^2 v}{4\pi^2 \epsilon_0 a^2}\left(-\hat{k}\right) \quad \left[\because \hat{j} \times \hat{i} = -\hat{k}\right]$$

$$\therefore S = -\frac{\lambda^2 v}{4\pi^2 \epsilon_0 a^2}\hat{k}$$

Radiation Pressure and Momentum of Electromagnetic Wave:-

Momentum:- The electromagnetic wave also carries linear momentum with it

The linear momentum carried by the portion of wave having energy U is given by:-

$E = mC^2$

$E = m \times C \times C$

$$E = p \times C \Rightarrow p = \frac{E}{C} \Rightarrow \boxed{p = \frac{\text{Energy (U)}}{C}}$$

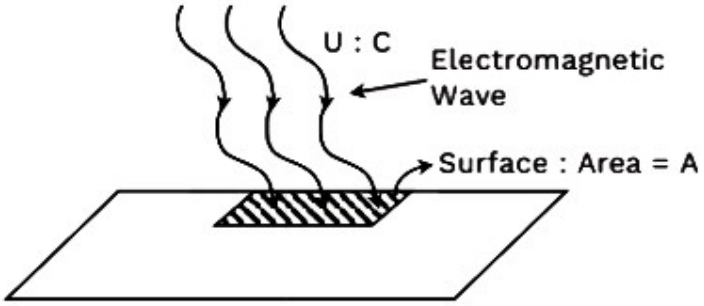
Where p = Momentum of EMW

U = Energy

C = Velocity of light

Radiation Pressure :- Electromagnetic waves carry energy and momentum on electromagnetic wave also exerts pressure called radiation pressure.

Case 1 : For completely absorbing surface :-



$$\text{Momentum (p)} = \frac{U}{C}$$

Due to momentum exist in EM wave. EM wave exert a force or surface.

$$\vec{F} = \frac{\overline{\Delta p}}{\Delta t} = \left(\frac{O - \vec{P}}{\Delta t}\right) = \frac{-U}{t \times C}$$

$$\boxed{|\vec{F}| = \frac{U}{C \times t}}$$

Due to this force ; surface experience a radiation pressure

$$P = \frac{|\vec{F}|}{A} = \frac{U}{A \times C \times t}$$

$$\boxed{P = \frac{U}{A \times C \times t}}$$

We know ; $I = \frac{U}{A \times t}$

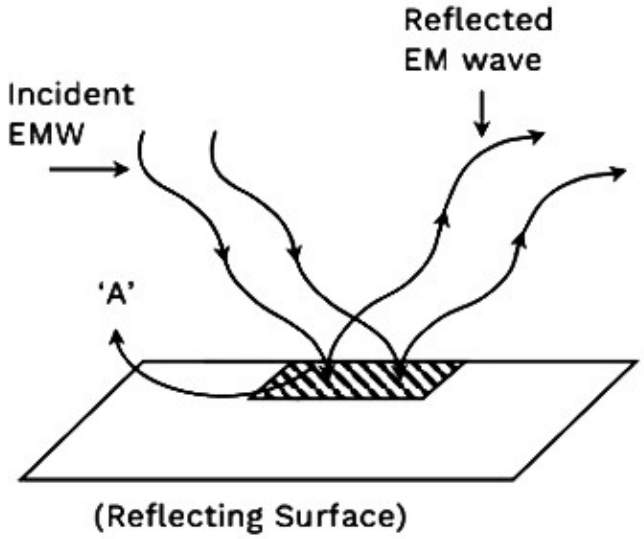
So, radiation pressure $\boxed{P = \frac{I}{C}}$

Where : P = Radiation Pressure

I = Intensity of EMW

C = Velocity of EMW (Light)

Case 2: For complete reflecting surface :



$$F = \frac{\overline{\Delta P}}{\Delta t} = \frac{\overline{P_f} - \overline{P_i}}{\Delta t}$$

$$\vec{F} = \frac{-P - (+P)}{\Delta t}$$

$$\left|\vec{F}\right| = \left|-\frac{2P}{\Delta t}\right| \qquad ...(1)$$

From relation, momentum of EMW is given by

$$P = \frac{U}{C} - 2 \qquad ...(2)$$

From equation (1) and (2)

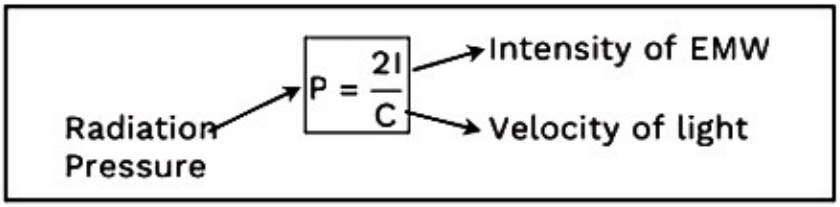
Force exerted on the surface p → $\boxed{\left|\vec{F}\right| = \frac{2U}{t \times C}}$

Radiation Pressure on the surface

$$P = \frac{F}{A} = \frac{2 \times U}{A \times t \times C}$$

$$= \frac{1}{C} \times 2\left(\frac{U}{A \times t}\right)$$

$$= \frac{2}{C} \text{ (Energy flow per unit area per unit time)}$$



Q. **An EM wave of intensity/falls on a surface kept in vacuum and exerts radiation pressure p on it. Which of the following are true?**

(A) Radiation pressure is $\frac{I}{C}$ if the wave is totally absorbed.

(B) Radiation pressure is $\frac{I}{C}$ if the wave is totally reflected.

(C) Radiation pressure is $\frac{2I}{C}$ if the wave is totally reflected.

(D) Radiation pressure is in the range $\frac{I}{C} < p > \frac{2I}{C}$ for real surface.

Sol. **Ans.** (A, C, D)

Radiation pressure (p) lies between I/C and $\frac{2I}{C}$ for real surfaces.

When wave is fully absorbed,

Radiation Pressure $(P) = \frac{\Delta I}{C} = \frac{I-0}{C} = \frac{I}{C}$ (When wave is fully absorbed by surface, the momentum of reflected wave per unit time per Area = 0)

When wave is fully reflected, momentum per unit time per unit is $= \frac{-I}{C}$

Radiation pressure $(p) = \frac{I}{C} - \left(\frac{-I}{C}\right)$

$$= \frac{2I}{C}$$

Q. **Prove that the radiation pressure exerted by an EM wave of intensity I on a surface kept in Vacuum is $\frac{I}{C}$.**

Sol. $\text{Pressure} = \frac{\text{Force}}{\text{Area}}$

$$\text{Force} = \frac{\text{Change in moementum}}{\text{Change in time}} = \frac{\Delta p}{\Delta t}$$

$$\therefore \quad P = \frac{\Delta p / \Delta t}{A} \times \frac{C}{C} \Rightarrow p = \frac{1}{AC \times \Delta t} \times C \times \Delta p = \frac{1}{AC\Delta t} \times \Delta U$$

$(\because \quad C\Delta p = U)$

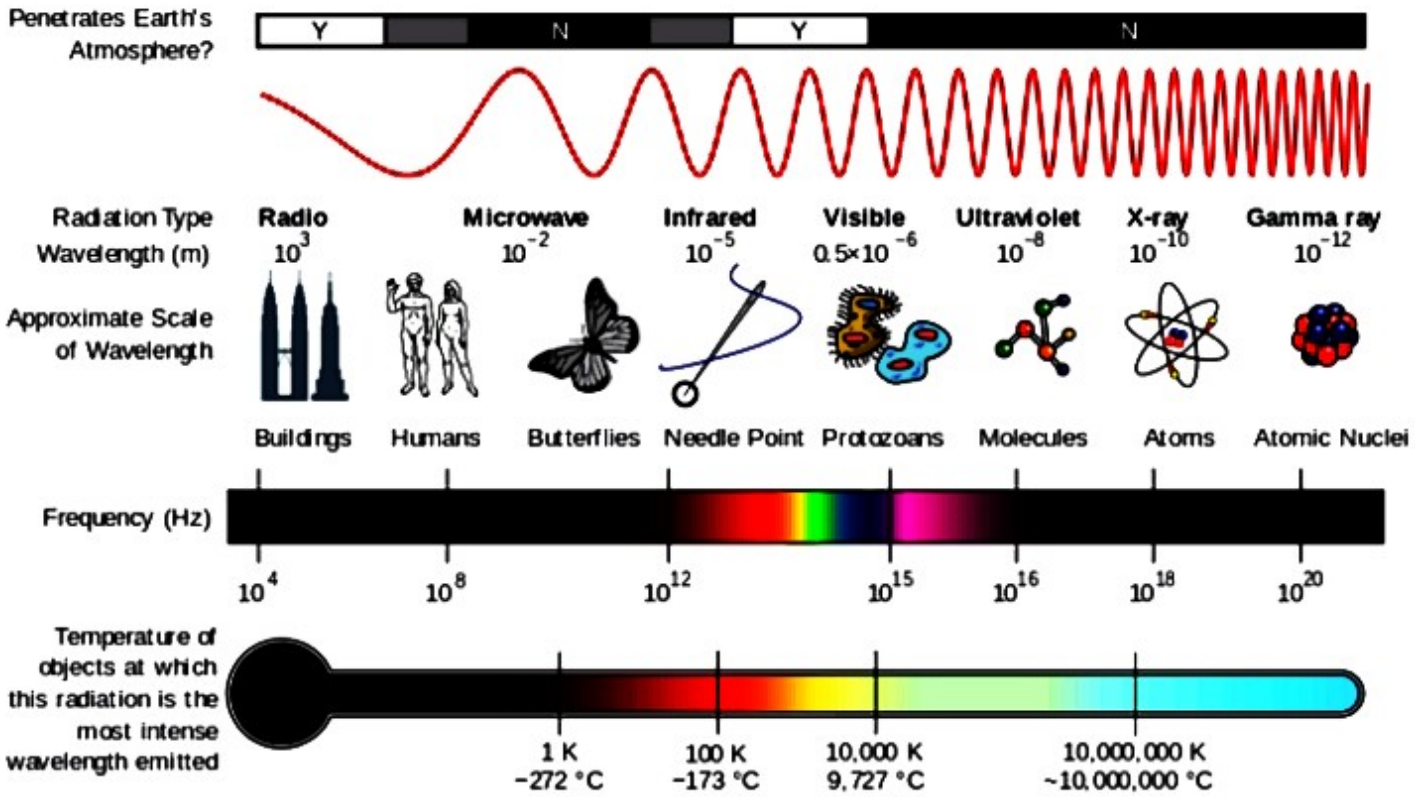
$$\text{Intensity, } I = \frac{\Delta U}{A \times \Delta t} = P \times C$$

$$\therefore \quad \boxed{P = \frac{I}{C}}$$

Electromagnetic Spectrum :-

At the time maxwell predicted the existence of electromagnetic waves, the only familiar electromagnetic waves were the visible light waves. The existence of ultraviolet and infrared waves was barely established. By the end of the nineteenth century, x-rays and gamma rays had also been discovered. We now know that electromagnetic waves include visible light waves, x-rays, gamma rays, radio waves, microwaves, ultraviolet, and infrared waves.

The classification of the electromagnetic wave according to frequency is the electromagnetic spectrum (shown in the figure).



	Rays	Wavelength	Frequency	Production	Properties Range	Application
1.	Gamma Rays	<0.01A°>	>3×10^{20}Hz	Nuclear decay	(a) High Penetrating Power (b) Uncharged (c) Low ionising Power	(a) Gives information on Nuclear Structure (b) Medical treatment etc.
2.	X-Rays	10A° – 0.01A°	3×10^{17}Hz to 3×10^{20}Hz	Inner electronic transitions and fast collisions	(a) Low Penetrating Power. (b) Other Properties Similar to γ-Rays except Wavelength	(a) Medical Diagnosis and treatment (b) Study of Crystal Structure (C) Industrial Radiography
3.	Ultraviolet rays	4000A° to 10A°	7.5×10^{14}Hz to 3×10^{10}Hz	Thermal agitation and electronic transitions	(a) All Properties of Light (b) Photo Electric Effect	(a) To detect adulteration, writing and signature (b) Sterilization of water due to its destructive action on Bacteria.
4.	Visible Light	4000A° to 7000A°	4.3×10^{14}Hz to 7.5×10^{14}Hz	Thermal agitation and electronic transitions	(a) Sensitive to Huma Eye	(a) To sec. objects (b) To study molecular structure.

Rays		Wavelength	Frequency	Production	Properties Range	Application
5.	Infra-red Rays	1mm to 7000A°	3×10^{11}Hz to 4.3×10^{14}Hz	Thermal agitation and electronic transitions	(a) Thermal Effect (b) All Properties Similar to those of Light except 'λ'	(a) Used in industry medicine and astronomy (b) Used for fog or Haze Photography (c) Elucidating molecular structure
6.	Micro-Waves	0.1m to 1mm	3×10^{9}Hz to 3×10^{11}Hz	Accelerating charges and thermal agitation	(a) Phenomena of reflection, refraction and diffraction	(a) Radios and Telecommunication (b) Analysis of fined details of molecular structure.
7.	Radio Wave	0.3 to few kms	$<3\times10^{9}$Hz	Accelerating charges	(a) Exhibit waves like properties more than particle like properties.	(a) Radio Communication

Frequency		Wavelength	Frequency	Application
(A)	Super High Frequency (SHF)	0.01 to 0.1 m	3×10^{10} Hz to 3×10^{9}Hz	
(B)	Ultra High Frequency (UHF)	0.01m to 1m	3×10^{9}Hz to 3×10^{8}Hz	Radio, Radio, and Satellite Communication (Micro Waves) Radar and Television broad cast short distance communication Television Communication
(C)	Very High Frequency (VHF)	1m to 10m	3×10^{8}Hz to 3×10^{7}Hz	
(D)	High Frequency (HF)	10m to 100m	3×10^{7}Hz to 3×10^{6}Hz	
(E)	Medium Frequency (MF)	100m to 1000m	3×10^{6}Hz to 3×10^{5}Hz	Medium distance Communication Telephone Communication, Marine and navigation use, long range Communication. Long distance communication.
(F)	Low Frequency (LF)	1000m to 10000m	3×10^{5}Hz to 3×10^{4}Hz	
(G)	Very Low Frequency (VLF)	10000m to 30000m	3×10^{4}Hz to 3×10^{4}Hz	

ATMOSPHERE :-

The atmosphere of the earth extends about 300 km. The composition of the atmosphere is as given below.

1. **Troposphere :-** Troposphere extends up to the light of about 12 km. This layer consists of water droplets, Vapor, dust particles, etc. The density of air decreases as we move from the bottom to the top of this layer. This layer is responsible for all the weather phenomena affecting our environment.

2. **Mesosphere :-** The mesosphere extends from about 50km to about 80 km.

3. **Ionosphere :-**The atmosphere above the mesosphere is called the ionosphere. This layer is composed partly of electrons and positive Ion. The rest of the atmosphere is composed mostly of neutral molecules.

The atmosphere is transparent to visible radiation, and we can see the sun and the stars through it clearly—however, most infrared radiation is not able to pass through, as it is absorbed by the atmosphere. Low lying clouds in the atmosphere also prevent infrared radiation from passing through the ozone layer blocks the passage of ultraviolet radiation from the sun.

Q. **TThis question has statement-1 and statement-2 of the four choices given after the statement; choose the one that best describes the two statements.**

Statement -1 : out of radio waves and microwaves, the radio waves under more diffraction.

Statement -2 : Radio waves have greater frequency compared to micro-waves.

(a) **Statement-1 is true, statement-2 is false.**

(b) **Statement-1 is false, statement-2 is true.**

(c) **Both statement is true and statement-2 is correct explanation of statement-1.**

(d) **Both statement is true and statement-2 is not correct explanation of statement-1**

Sol. **Ans.** (A)

Q. **Match List-I (Wavelength range of electromagnetic spectrum) with List-II. (method of production of these waves) and select the correct option from the options given below the lists**

(a)	700 nm to 1mm	(i)	Vibration of atoms and molecules
(b)	1nm to 400 nm	(ii)	Inner shell electrons in atoms moving from one level to a lower level
(c)	$< 10^{-3}$ nm	(iii)	Radioactive decay of the molecules
(d)	1mm to 0.1 m	(iv)	Magnetron valve

(a) (a)–(i), (b)–(ii), (c)–(iii), (d)–(iv) **(b) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)**
(c) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i) **(d) (a)–(ii), (b)–(iii), (c)–(iv), (d)–(i)**

Sol. **Ans.** (A)
(a)-(i), b(ii), (c)-(iii), (d)-(iv)

(a) Waves of 700 nm to 1mm can be produced by vibration of atoms and molecules.
(b) Waves of 1 nm to 400 mm are produced when an electron moves from one energy level to lower level in an atom.
(c) Waves of wavelength <10 – 3 nm can be produced by radioactive decay of the nucleus.
(d) Waves of 1 mm to 0.1m are produced by magnetron valve."

Q. **The AM range of Radio waves have frequency –**
(A) Less than 30 MHz
(B) More than 30 MHz
(C) Less than 20,000 Hz
(D) More than 20,000 Hz

Sol. **Ans.** (A)

Q. **Which of the following is/are not used as diagnostic tool in medicine?**
(A) X-Ray
(B) Y-Ray
(C) Ultrasonic wave
(d) None wave.

Sol. **Ans.** (B)

Q. **Earth satellite use the detectors of which electromagnetic waves?**
(A) I R – Waves
(B) X – Rays
(C) UV – Rays
(D) Radio – Waves

Sol. **Ans.** (A)

Q. **One can not get turned or sunburn through glass window because**
(A) Glass absorbs UV radiation.
(B) Glass reflects UV radiation.
(C) Glass is transparent to UV radiation.
(D) None of these

Sol. **Ans.** (A)
Glass is transparent because it lets all the colors go true it but what we can not see is that it acts as a filter for UV light. Most of UV radiation is stopped by glass & this is why you will not get sunburns behind a glass.